ALGEBRA 2 · RATIONAL FUNCTIONS

# Asymptote & Hole Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Asymptote & Hole Explorer** on Mr. K's Classroom: Algebra 2 → Rational Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict before you touch the tool

1. $f(x) = \frac{(x-2)(x+1)}{(x-2)}$. What do you think happens at $x = 2$?

## 2 Explore use the tool and record what you see

### Build each function

Use the factor switches and sliders.

| Function | Holes | Vertical asymptotes | Horizontal asymptote | x-intercepts |
|---|---|---|---|---|
| $\frac{(x+2)}{(x-2)}$ | | | | |
| $\frac{(x-2)(x+1)}{(x-2)}$ | | | | |
| $\frac{1}{((x-1)(x+4))}$ | | | | |
| $\frac{3(x+1)}{(x-3)}$ | | | | |
| $\frac{(x+3)(x-1)}{((x-1)(x-5))}$ | | | | |

## 3 Write the rule fill in each blank from what you observed

1. A factor that cancels from the top and bottom makes a _____.
2. A factor left in the denominator makes a _____.
3. Top degree < bottom degree: horizontal asymptote _____. Equal degrees: y = _____. Top degree > bottom degree: _____.
4. The x-intercepts come from the factors left in the _____.

## 4 Check no tool: show your work

1. Find the hole, asymptotes and x-intercept of $f(x) = \frac{(x+4)(x-2)}{((x-2)(x+1))}$.

2. Find the horizontal asymptote of $\frac{5x^2}{(x^2 - 9)}$.

3. Find the vertical asymptotes of $\frac{2}{(x^2 - 16)}$.

## 5 Reflect

Why is a hole different from a vertical asymptote?